In any problem that deals with uncertainty, we will, quite obviously, never be able to make a definitive statement. However, if the problem is well defined, we should be able to list all the possible outcomes. If an uncertain event is repeated often enough, the frequency of the outcomes should reflect the probability of the different possible outcomes. The difficulty arises when we are concerned with an event that happens only once.

How do we estimate the probability of passing tomorrow's science test or the probability of the Green Bay Packers winning the Super Bowl? The problem we face is that each of these events is unique. We can look back at all the statistics on the Green Bay games, but we don't have enough exact matchups with the exact personnel who played each other repeatedly under identical circumstances. We can recall previous science exams to get an idea of how well we test, but all tests are not identical and our knowledge is not constant.

Without repeated tests that would produce a frequency distribution, how can we calculate probability? We cannot. Instead we must rely on subjective interpretation of probabilities. We do it all the time. We might say the odds of the Packers' winning the Lombardi trophy are two to one, or the possibility of passing that difficult science test is one in 10. These are probabilistic statements; they describe our *degree of belief* about the event. When it isn't possible to do enough repetitions of a certain event to get an interpretation of probability based on frequency, we have to rely on our own good sense.

You can immediately see that many subjective interpretations of those two events would lead you in the wrong direction. In subjective probability, the burden is on you to analyze your assumptions. Stop and think them through. Are you assuming a one-in-10 chance of doing well on the science test because the material is more difficult and you haven't adequately prepared, or because of false modesty? Is your lifelong loyalty to the Packers blinding you to the superior strength of the other team?

According to the textbooks on Bayesian analysis, if you believe that your assumptions are reasonable, it is "perfectly acceptable" to make your subjective probability of a certain event equal to a frequency probability.[12] What you have to do is sift out the unreasonable and illogical and retain the reasonable. It is helpful to think about subjective probabilities as nothing